

SAENZ PENA, Argentina

WMO: 871480

Lat: 26.7464S

Lon: 60.4867W

Elev: 93

StdP: 100.21

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	3.3	-3.0	3.0	8.8	-1.1	3.5	9.3	12.3	22.8	11.7	25.3	0.7	230	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.4	38.1	24.8	36.7	24.9	35.4	24.5	27.3	33.9	26.7	33.1	26.1	32.3	5.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.2	30.6	25.1	20.4	29.9	24.5	19.7	29.3	87.6	34.0	84.5	33.4	82.0	32.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	9.1	7.8	-3.1	41.1	2.3	1.5	-4.7	42.1	-6.1	43.0	-7.4	43.8	-9.0	44.9		
			-3.0	28.7	2.5	1.5	-4.8	29.8	-6.3	30.6	-7.7	31.5	-9.5	32.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	27.4	26.6	24.5	21.8	18.2	16.1	15.8	17.7	20.2	22.9	24.4	26.5
	DBStd	5.92	2.67	3.30	3.22	4.14	4.69	5.07	5.62	5.89	5.34	4.25	3.42	3.16
	HDD10.0	20	0	0	0	0	1	5	9	4	1	0	0	0
	HDD18.3	439	0	0	1	14	62	104	122	88	39	8	1	0
	CDD10.0	4334	539	464	449	354	257	189	189	242	305	401	432	512
	CDD18.3	1710	281	231	192	118	59	38	42	67	94	151	184	254
	CDH23.3	17972	3306	2506	1813	971	413	224	348	776	1072	1584	1991	2968
(13)	CDH26.7	8456	1704	1226	801	364	125	52	118	364	509	746	933	1513
(14)	Wind	WSAvg	3.1	2.8	2.8	2.5	2.6	2.9	3.2	3.7	3.9	3.9	3.4	3.0
(15) Precipitation	PrecAvg	1054	147	132	146	141	54	25	21	24	41	77	129	130
	PrecMax	1729	402	291	397	362	185	100	91	58	129	242	242	304
	PrecMin	643	18	23	24	20	1	0	0	0	0	17	16	18
	PrecStd	254	80	78	82	83	43	25	20	20	33	48	61	71
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	38.6	37.5	34.4	32.7	30.1	32.1	36.1	38.5	39.4	38.6	39.3
		MCWB	25.4	25.5	24.5	24.3	23.9	22.3	20.4	22.5	22.8	24.4	25.2	25.0
	2%	DB	37.0	36.7	35.1	32.7	29.9	28.3	29.9	33.8	35.6	36.3	35.9	37.0
		MCWB	25.4	25.6	24.4	24.3	22.6	21.8	20.7	21.5	22.0	24.5	24.5	25.4
	5%	DB	35.6	34.9	33.1	30.8	27.5	26.2	27.5	31.1	32.4	33.4	33.8	35.4
		MCWB	25.1	25.3	24.1	23.4	21.9	21.1	20.0	21.0	21.2	23.1	23.6	24.9
	10%	DB	34.0	33.0	31.1	28.8	25.3	23.7	24.6	27.8	29.3	30.8	31.8	33.6
		MCWB	24.9	24.8	23.6	22.7	20.9	19.9	18.5	19.5	19.8	21.8	22.7	24.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	28.1	27.5	26.6	25.1	23.7	23.0	24.0	24.6	26.6	27.4	27.7
		MCDB	34.2	34.2	33.3	32.0	29.7	27.7	29.0	33.5	34.2	34.4	34.8	34.9
	2%	WB	27.2	27.1	26.3	25.6	23.9	22.5	21.5	22.4	23.4	25.4	26.1	26.8
		MCDB	33.6	33.4	31.7	30.3	28.3	26.5	27.6	31.6	31.6	33.5	33.0	33.7
	5%	WB	26.5	26.2	25.4	24.7	22.9	21.5	20.4	21.1	22.2	24.3	25.0	26.2
		MCDB	32.7	32.3	30.6	29.1	26.4	25.3	26.0	29.8	30.1	31.9	31.5	32.8
	10%	WB	25.8	25.5	24.6	23.7	21.7	20.4	19.3	19.8	21.0	23.2	24.1	25.5
		MCDB	31.6	31.1	29.2	27.3	24.5	23.6	24.3	26.7	27.9	29.6	29.6	31.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	10.7	11.0	10.2	10.4	10.4	12.2	13.5	13.3	11.7	12.0	11.6
		MCDBR	13.7	13.0	13.8	12.3	11.9	11.2	13.7	16.1	16.5	15.3	15.1	13.9
		MCWBR	4.5	4.2	5.2	5.1	5.5	5.4	6.0	6.4	6.6	5.9	5.8	4.6
	5% WB	MCDBR	11.0	10.7	11.0	9.9	9.4	9.9	12.3	15.2	13.7	13.2	12.0	11.3
		MCWBR	4.5	4.3	5.1	4.8	4.7	5.2	6.2	6.7	6.1	5.9	5.4	4.4
(40) Clear-Sky Solar Irradiance	taub		0.408	0.405	0.393	0.395	0.369	0.375	0.370	0.462	0.539	0.476	0.403	0.408
	taud		2.391	2.407	2.429	2.391	2.424	2.397	2.388	2.090	1.889	2.138	2.363	2.379
	Ebn at Noon		936	923	903	850	829	797	818	770	756	847	934	938
	Edh at Noon		129	124	115	109	96	94	98	145	194	160	131	131
(44) All-Sky Solar Radiation	RadAvg		6.75	6.13	5.28	4.04	3.23	2.70	3.25	4.04	4.82	5.56	6.52	6.69
	RadStd		0.51	0.40	0.35	0.44	0.39	0.27	0.26	0.37	0.56	0.55	0.46	0.47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page